

ICLR Publication Plan: CARE-Bench and CARE-Graph

Generated from jitai-autism-behavior-video-reasoning/ICLR_PUBLICATION_PLAN.md. Hyperlinks in the table of contents jump to sections in digital PDF viewers.

Table of Contents

ICLR Publication Plan: CARE-Bench and CARE-Graph

Executive Decision

Sources Consulted

Five ICLR Program-Chair Feedback Loops

Loop 1: Scope and Claim

Loop 2: Novelty

Loop 3: Dataset

Loop 4: Method

Loop 5: ICLR Positioning

Final Publication Plan

1. Problem Formulation

2. Motivation

3. Contribution Claims

4. Related Work Comparison

5. Novelty

6. Dataset

7. Method

8. Baselines

9. Evaluation Metrics

10. Experiments

11. Failure Taxonomy

12. Step-by-Step How-To

13. Paper Outline

14. Reviewer-Proof Abstract

15. Minimum Submission Checklist

16. Final Risk Register

Final Standard

Generated: May 20, 2026

This document executes `TASK.md`: it studies the local research reports and current plan, cross-checks the referenced literature and project pages, runs five harsh ICLR program-chair feedback loops, and turns the project into a concrete publication plan.

Executive Decision

The strongest ICLR version is not "an agentic system for autism-relevant behavior video reasoning." That framing is too broad, ethically exposed, and technically hard to falsify.

The strongest version is:

CARE-Bench and CARE-Graph: evidence-calibrated cue-access-response reasoning for participant-centered social video understanding.

The hard technical object is an InteractionEpisode graph:

```
observable cue -> recipient sensory access -> observable response/no-response -> latency -> evidence spans -  
> claim level
```

The paper should argue that current video-language models can caption social scenes fluently while failing to recover interaction mechanics: who cued whom, whether the recipient had observable access, what response followed, how long it took, what evidence supports the claim, and what cannot be inferred.

The top-1% ICLR path requires a benchmark plus a method. A benchmark-only paper may be publishable if the data is excellent, but it is unlikely to feel like a top-1% ICLR submission. The method should be a constrained evidence-graph inference system with a trainable edge scorer and a claim verifier, not just an LLM prompt wrapper.

Sources Consulted

Local sources:

- `knowledge/multi-char-social-agent.structured.md`
- `knowledge/multimodal-fusion-agent.structured.md`
- `knowledge/multimodal-fusion-agent.md`
- `plan-05-20.md`

External publications and project pages:

- Read the Room / R3-Bench and R3-FDT, ICLR 2026 OpenReview: <https://openreview.net/forum?id=TJilJnZjpw>
- SIV-Bench project page: <https://kfq20.github.io/sivbench/>
- Social Genome: <https://arxiv.org/abs/2502.15109>
- MOMENTS: <https://arxiv.org/abs/2507.04415>

- EGAgent / Agentic Very Long Video Understanding: <https://arxiv.org/abs/2601.18157>
- VideoExplorer / VideoDeepResearch code page: <https://github.com/RUC-NLPIR/VideoDeepResearch>
- VideoMultiAgents: <https://arxiv.org/abs/2504.20091>
- NA-VQA: <https://arxiv.org/abs/2603.19481>
- Social-IQ 2.0 challenge: <https://cmu-multicomp-lab.github.io/social-iq-2.0/>
- MovieGraphs: <https://moviegraphs.cs.toronto.edu/>
- Ego4D: <https://arxiv.org/abs/2110.07058>
- Ego-Exo4D: <https://arxiv.org/abs/2311.18259>
- LongVideoBench: <https://arxiv.org/abs/2407.15754>
- MLVU: <https://arxiv.org/abs/2406.04264>
- OmniBench: <https://arxiv.org/abs/2409.15272>

Five ICLR Program-Chair Feedback Loops

Loop 1: Scope and Claim

Program-chair critique:

The current plan is intellectually rich but still tries to be a research program. It wants social reasoning, long-video segmentation, multimodal fusion, behavior chains, agent orchestration, ethics, and clinical-adjacent motivation. Reviewers will ask: what exactly is the task, what exactly is new, and what exactly improves?

Refinement:

Narrow the paper to participant-centered interaction mechanics. The thesis is not "understand social behavior." The thesis is "recover observable cue-access-response episodes and avoid unsupported inferences." Keep ASD relevance as a motivation for why conservative, observable reasoning matters, but do not make ASD a claim or dataset axis.

Concrete change:

The first paragraph of the paper should define the gap:

A caption says what happened in a scene. An InteractionEpisode says who cued whom, whether the recipient had observable access, what response followed or did not follow, when it happened, and what evidence supports that claim.

Loop 2: Novelty

Program-chair critique:

CARE-Bench as written risks looking like a relabeling of R3-Bench, SIV-Bench, MOMENTS, Social Genome, and Social-IQ 2.0. Those already evaluate social state, ToM, grounded social reasoning, and social interaction. What is the new scientific object?

Refinement:

The novelty is not another social QA benchmark. The novelty is an observable interaction graph with access and overclaim constraints. R3-Bench focuses on mental-physical causal chains, SIV-Bench splits scene/state/dynamics, MOMENTS targets ToM categories, and Social Genome scores reasoning traces. CARE should instead test whether a model can recover interaction mechanics without pretending to know hidden mental states.

Concrete change:

Frame CARE as "pre-ToM" or "evidence-before-ToM." It is the layer needed before credible mental-state reasoning:

Before asking what P2 believed or intended, the system must establish whether P2 could see or hear P1's cue and whether P2 produced an observable response.

Loop 3: Dataset

Program-chair critique:

The dataset plan is still under-specified. If the dataset is mostly scraped public video, reviewers will attack licensing, consent, cultural bias, child/privacy risk, and annotation ambiguity. If the dataset is too small, they will call it a pilot. If it is too synthetic, they will call it brittle.

Refinement:

Use a two-track dataset strategy:

1. A public/research-source seed set for broad comparability and fast baseline work.
2. A purpose-collected adult dyad/triad dataset designed specifically for cue-access-response ambiguity.

Do not use children, clinical sessions, or diagnostic framing in the main ICLR submission. That belongs in a future IRB-controlled translational study.

Concrete change:

The ICLR benchmark should target 300-500 clips and 1,000-2,000 adjudicated InteractionEpisodes. The purpose-collected slice should be the differentiator: it lets you intentionally control access ambiguity, simultaneous cues, offscreen cues, no-response cases, and subtle object-mediated responses.

Loop 4: Method

Program-chair critique:

"Evidence graph + verifier" sounds plausible but generic. Every 2025-2026 video-agent paper has retrieval, graphs, or verification. EGAgent already uses entity scene graphs and search; VideoExplorer already iterates over subquestions and temporal grounding; VideoMultiAgents already combines vision, scene graphs, and text agents. What does this method do that those do not?

Refinement:

CARE-Graph must be explicitly access-calibrated and episode-constrained. Its inference target is not general answer generation but a typed graph with legal and illegal claims. The method should include:

- candidate event proposal from VLM/audio/pose/transcript tools

- participant identity and speaker alignment
- typed candidate edges for cue, access, response, and evidence support
- a trainable edge scorer for cue-response and access edges
- a global constrained decoder or integer program that selects a consistent episode graph
- a claim-level verifier that rejects unsupported mental-state, trait, and clinical claims

Concrete change:

Make the method contribution:

CARE-Graph: a constrained, access-aware episode inference model over multimodal evidence proposals.

The important ablation is not "agent vs no agent." It is:

flat VLM -> schema-prompted VLM -> evidence cards -> unconstrained graph -> constrained graph -> constrained graph + verifier

Loop 5: ICLR Positioning

Program-chair critique:

Top ICLR papers usually either expose a fundamental failure mode with unusually clean evaluation or introduce a method with broad implications. CARE has both potential, but the writing must be ruthless. If the paper reads like "we made a useful benchmark for social video," it will be borderline. If it reads like "social video reasoning is impossible to evaluate until we separate observable access from hidden mental-state inference," it becomes a sharper scientific claim.

Refinement:

The paper should be positioned around a conceptual fault line:

Current social-video benchmarks reward plausible social interpretation. CARE tests whether interpretation is evidentially licensed.

Concrete change:

Use three headline results:

1. Strong VLMs produce fluent but structurally wrong social descriptions.
2. Oracle perception helps but does not solve access/response linking, showing this is a reasoning problem.
3. CARE-Graph improves episode recovery and sharply reduces overclaiming while preserving useful specificity.

Final PC verdict after five loops:

Weak version: benchmark with interesting labels, likely rejected or borderline.

Strong version: a falsifiable benchmark plus constrained inference method that reveals a hidden failure mode in social video reasoning, plausible ICLR.

Top-1% version: purpose-collected, ethically clean data; strong open and frontier baselines; human ambiguity analysis; clear method gains on cue-response linking and overclaim reduction; and writing that keeps every claim observable.

Final Publication Plan

1. Problem Formulation

Define participant-centered social video reasoning as structured episode extraction from a video clip.

Input:

video V, optional audio A, optional transcript T, question or extraction prompt Q

Output:

```
{
  "participants": [
    {
      "participant_id": "P1",
      "visible_spans": [[12.4, 38.1]],
      "speaker_spans": [[13.2, 14.8]],
      "track_confidence": "high"
    }
  ],
  "interaction_episodes": [
    {
      "episode_id": "IE_001",
      "cue_event": {
        "event_id": "C001",
        "start_sec": 13.2,
        "end_sec": 14.8,
        "actor_id": "P1",
        "cue_type": "speech_gesture",
        "candidate_recipient_ids": ["P2"],
        "evidence_spans": [[13.2, 14.8]]
      },
      "recipient_access": {
        "recipient_id": "P2",
        "access_label": "visible_and_audible",
        "access_evidence": ["P2 visible", "P1 speech audible", "P2 head/body orientation compatible with access"],
        "forbidden_inference": ["P2 understood", "P2 intended to comply"]
      },
      "response_event": {
        "event_id": "R001",
        "start_sec": 15.1,
        "end_sec": 17.0,
        "actor_id": "P2",
        "response_label": "possible_response",
        "response_type": "object_action_response",
        "latency_sec": 0.3
      },
      "claim_level": "observable_or_weak_inference",
      "unsupported_claims": []
    }
  ]
}
```

```
}  
}
```

Core distinction:

- Observable: P2 turns toward the toy after P1 points.
- Weak evidence-grounded inference: P2 may be responding to P1's gesture.
- Unsupported mental state: P2 understands the request.
- Disallowed trait/clinical inference: P2 is rude, impaired, autistic, inattentive.

2. Motivation

Social video benchmarks are getting harder, but many still reward plausible social interpretation. R3-Bench and MOMENTS focus on mental states and ToM; SIV-Bench evaluates scene, state, and dynamics; Social Genome evaluates grounded reasoning traces. These are important, but they do not isolate a prerequisite: whether the model can reconstruct the observable mechanics that license any later social interpretation.

CARE targets this missing layer. It matters because:

- Current MLLMs can describe a scene and still swap participant roles.
- They can infer intent while missing whether the recipient could see or hear the cue.
- They often treat temporal adjacency as causality.
- They overclaim hidden attention, comprehension, personality, and clinical meaning.
- In human-centered or clinical-adjacent settings, unsupported inference is not just a benchmark error; it is a safety failure.

The paper should make this moral-technically boring in the best way: no diagnosis, no hidden mental state labels as ground truth, no claims beyond evidence.

3. Contribution Claims

Make exactly five contribution claims.

1. Task formalization:

We formalize cue-access-response reasoning as InteractionEpisode extraction, separating observable evidence from weak inference and unsupported claims.

1. Dataset:

We introduce CARE-Bench, a benchmark of naturalistic adult social interaction clips labeled with participants, cue events, access labels, response/no-response events, latency, evidence spans, ambiguity, and overclaim categories.

1. Metrics:

We define component, graph, grounding, calibration, and overclaim metrics for participant-centered social video reasoning.

1. Method:

We propose CARE-Graph, a constrained evidence-graph inference system with a trainable edge scorer and claim verifier for producing valid InteractionEpisodes.

1. Empirical finding:

We show that strong VLMs remain weak at access reasoning, cue-response linking, and overclaim control, and that constrained evidence-graph inference improves these dimensions over flat and schema-prompted baselines.

4. Related Work Comparison

Line of work	What it contributes	Why CARE is different
R3-Bench / Read the Room	Mental-state variables and mental-physical causal chains in complex social scenarios	CARE avoids hidden mental-state supervision and tests whether observable access/response mechanics are recovered first
SIV-Bench	Broad social interaction evaluation across scene, state, and dynamics	CARE is not broad social QA; it is typed episode extraction with evidence and overclaim constraints
Social Genome	Human reasoning traces grounded in visual, verbal, vocal, and external knowledge cues	CARE scores structured interaction mechanics, access, latency, and illegal claim control
MOMENTS	Multimodal ToM over narrative-rich short films	CARE is pre-ToM and evidence-calibrated rather than mental-state centered
Social-IQ 2.0	Social video QA with multimodal inputs	CARE avoids multiple-choice shortcut bias and requires explicit evidence spans and graph outputs
MovieGraphs	Human-centric graph annotations in movie clips	CARE brings graph structure to observable interaction episodes in releasable/consented social videos
EGAgent	Entity scene graphs and structured search for very long egocentric video	CARE uses graphs for access-calibrated episode inference, not general long-video QA
VideoExplorer	Iterative planning, temporal grounding, and scalable perception	CARE's novelty is the typed social evidence target and claim legality constraints
VideoMultiAgents	Specialized vision, scene-graph, and text agents for VideoQA	CARE is not agent collaboration; it is constrained participant-centered episode inference
NA-VQA	Long-range narrative reasoning across movies	CARE focuses on short-to-medium social interaction mechanics and observable evidence rather than full-movie narrative causality

5. Novelty

The paper's novelty should be stated precisely:

CARE is the first benchmark and inference framework that evaluates whether multimodal models can recover evidence-licensed cue-access-response episodes from social videos while explicitly penalizing unsupported social, trait, and clinical claims.

Do not claim:

- "solves social understanding"
- "understands autism"
- "detects attention"
- "infers intention"
- "diagnoses behavior"

Do claim:

- "recovers observable interaction mechanics"
- "separates evidence from interpretation"
- "measures access and response ambiguity"
- "penalizes overclaiming"
- "finds a failure mode hidden by broad captioning and multiple-choice QA"

6. Dataset

6.1 Dataset Name

Use:

CARE-Bench: Cue-Access-Response Evaluation for Social Video Reasoning

Avoid overloading "clinical," "autism," or "behavior diagnosis" in the benchmark name.

6.2 Data Sources

Use three tiers.

Tier A: public/research-source seed set

- Ego4D/Ego-Exo4D clips where consent, license, and annotations support research use
- Social-IQ 2.0 clips where videos remain available and licenses permit
- human-robot interaction, collaborative task, teaching, or instructional interaction datasets with research-friendly use
- limited movie/TV benchmark examples only for comparison if redistribution is not needed

Tier B: purpose-collected adult interaction set

Collect adult dyads and triads under consent. Record short, naturalistic tasks:

- object handoff and object-mediated instruction
- collaborative puzzle or assembly
- teacher-learner demonstration
- peer explanation
- turn-taking conversation with visible gesture
- ambiguous/no-response interactions
- simultaneous cues from multiple initiators
- offscreen, occluded, audible-only, and visible-only cue conditions

Tier C: future translational extension

Separate from ICLR. If autism- or child-related data is ever used, it needs IRB, consent, expert annotation, and a different risk analysis. It should not be part of the main submission.

6.3 Scale

Pilot:

75 clips
200-300 InteractionEpisodes
3 annotators on 25 clips
expert adjudication

ICLR submission:

300-500 clips
1,000-2,000 InteractionEpisodes
100 clips purpose-collected
3 annotators on at least 30% of clips
fully adjudicated test set
domain-stratified splits

Stretch:

1,000+ clips
3,000-5,000 InteractionEpisodes
additional long-video and cross-cultural splits

6.4 Annotation Schema

Annotate:

- participant tracks and stable IDs
- speech turns and speaker identity when available
- cue event start/end
- cue actor
- candidate recipient
- cue type
- recipient access label
- response/no-response event
- response type
- cue-response link
- latency
- evidence spans
- ambiguity label
- unsupported claim categories

Access labels:

```
visible_and_audible
audible_only
visible_only
orientation_supports_access
orientation_unclear
occluded
offscreen
unknown
not_applicable
```

Response labels:

```
direct_response
possible_response
temporally_adjacent_behavior
no_observed_response
ambiguous_response
```

Claim levels:

```
0 observable
1 weak evidence-grounded inference
2 mental-state claim
3 trait/personality claim
4 clinical/diagnostic claim
```

6.5 Annotation Quality

Report:

- temporal IoU agreement for cue and response spans
- actor/recipient agreement
- access label macro agreement
- response label agreement
- latency agreement within tolerance
- Krippendorff alpha for categorical labels
- disagreement analysis by ambiguity stratum

Important: ambiguous clips are not annotation failures. They are part of the benchmark. Model scoring should reward calibrated uncertainty when humans also disagree.

7. Method

7.1 CARE-Graph Pipeline

```
video/audio/transcript
-> participant and speaker alignment
-> candidate event proposal
-> multimodal evidence graph
-> edge scoring
-> constrained episode decoding
```

```
-> claim verification
-> InteractionEpisode output
```

7.2 Evidence Graph

Nodes:

```
participant
speaker_turn
gesture_event
gaze_or_orientation_event
object_event
movement_event
candidate_cue
candidate_response
evidence_span
claim
```

Edges:

```
performs(participant, event)
speaks(participant, speaker_turn)
targets(candidate_cue, participant)
has_access(participant, cue)
precedes(cue, response)
possibly_causes(cue, response)
supports(evidence_span, claim)
contradicts(evidence_span, claim)
same_event(event, event)
```

7.3 Candidate Proposal

Use multimodal tools to propose candidates, but do not make them the contribution:

- VLM frame/clip descriptions
- transcript and diarization
- pose/head/body orientation when available
- object and hand interaction proposals
- shot or clip segmentation

Each proposal must carry:

```
timestamp
participant ID
modality
confidence
evidence span
source tool
```

7.4 Edge Scoring

Train lightweight edge scorers over graph candidates:

- cue-recipient edge score

- recipient-access edge score
- cue-response edge score
- response-label score
- evidence-support score

Features:

- temporal distance
- participant visibility
- audio audibility
- body/head orientation compatibility
- transcript target hints
- object continuity
- VLM semantic compatibility
- candidate confidence

This makes the method more than prompt engineering. The model learns when adjacency is insufficient and when access is uncertain.

7.5 Constrained Decoding

Decode the episode graph under constraints:

- cue must precede response
- response actor must be recipient
- access cannot be stronger than evidence
- no-response cannot coexist with a selected direct response
- a response can be linked to multiple cues only if ambiguity is flagged
- mental-state claims require explicit verbal evidence or are rejected
- trait and clinical claims are always rejected

Implement initially as integer linear programming, dynamic programming, or a constrained factor graph. If engineering time is short, use a deterministic constrained decoder over scored edges.

7.6 Claim Verifier

The verifier maps generated text or graph-derived explanations into claim levels. It rejects or hedges claims that exceed evidence.

Outputs:

```
accepted
accepted_with_hedge
needs_review
rejected
```

Verifier metrics must include both overclaim reduction and informativeness retention. A uselessly vague verifier is not a win.

8. Baselines

Use strong baselines. Weak baselines will sink the paper.

Flat baselines:

- video caption to parsed schema
- question-only or transcript-only controls when applicable
- sampled frames plus transcript

VLM baselines:

- strong open VLMs current at submission time
- strong proprietary VLMs where terms permit
- schema-prompted JSON output
- chain-of-thought or rationale-prompted variant if allowed by API and review policy

Agentic baselines:

- VideoExplorer-style temporal retrieval and zoom
- EGAgent-style entity graph retrieval if feasible
- VideoMultiAgents-style scene graph/text/vision decomposition

Oracle baselines:

- oracle clip segmentation
- oracle participant IDs
- oracle transcripts
- oracle cue proposals
- oracle access labels

Ablations:

- evidence cards only
- graph without trainable edge scorer
- graph without constraints
- graph without verifier
- verifier-only post-processing
- full CARE-Graph

9. Evaluation Metrics

Component metrics:

- Cue localization F1: temporal IoU plus actor match
- Cue type macro-F1
- Recipient accuracy
- Access macro-F1
- Response detection F1
- Response type macro-F1

- Cue-response pair F1
- Latency MAE and latency bucket accuracy
- Evidence grounding precision and recall
- Ambiguity calibration score

Graph metrics:

- InteractionEpisode exact match
- partial InteractionEpisode score
- episode graph node F1
- episode graph edge F1
- duplicate episode rate
- invalid graph rate

Claim metrics:

- overclaim violation rate
- severe overclaim rate
- mental-state overclaim rate
- clinical/trait claim violation rate
- verifier false rejection rate
- explanation informativeness score

Calibration metrics:

- abstention precision
- uncertainty expected calibration error
- selective prediction curve
- human-disagreement alignment

10. Experiments

Experiment 1: VLM failure diagnosis

Question:

Do strong VLMs recover participant-centered interaction mechanics?

Compare flat, schema-prompted, transcript-enhanced, and sampled-frame VLMs. Report component metrics and failure taxonomy.

Expected result:

High scene description quality, low cue-response pair F1, weak access labels, high overclaim rate.

Experiment 2: CARE-Graph main result

Compare:

structured VLM
evidence cards
unconstrained graph
constrained graph
constrained graph + verifier

Primary metrics:

- cue-response pair F1
- access macro-F1
- evidence grounding precision
- overclaim violation rate
- partial InteractionEpisode score

Experiment 3: Oracle bottleneck analysis

Use oracle participant IDs, cue proposals, segmentation, transcripts, and access labels.

Goal:

Separate perception failures from reasoning failures. If oracle perception still leaves weak cue-response linking, the paper has a stronger reasoning claim.

Experiment 4: Ambiguity and calibration

Stratify by:

- clear vs ambiguous recipient
- visible+audible vs one-modality-only access
- direct response vs no response
- simultaneous cues
- object-mediated response
- offscreen/occluded recipient

Reward calibrated uncertainty when human annotators disagree.

Experiment 5: Overclaim control

Compare generations before and after verifier.

Report:

- overclaim reduction
- severe overclaim reduction
- informativeness retention
- human preference for safe-but-useful explanations

Experiment 6: Data scaling and specialization

Train edge scorer with increasing labeled data:

50 clips -> 100 clips -> 250 clips -> 500 clips

Show whether purpose-collected access-ambiguous clips improve robustness more than generic public clips.

11. Failure Taxonomy

Use this taxonomy throughout the paper:

F1 missed cue
F2 wrong cue actor
F3 wrong recipient
F4 overstated access
F5 understated access
F6 missed response
F7 spurious response
F8 wrong cue-response link
F9 response latency error
F10 evidence span mismatch
F11 participant identity swap
F12 speech-speaker mismatch
F13 temporal adjacency mistaken for causality
F14 ambiguity treated as certainty
F15 unsupported mental-state claim
F16 trait/personality overclaim
F17 clinical/diagnostic overclaim
F18 duplicate episode from overlapping windows

12. Step-by-Step How-To

Week 1: Freeze the task

1. Freeze the InteractionEpisode schema.
2. Write annotation guidelines with positive and negative examples.
3. Define claim-level taxonomy.
4. Define the first version of metrics.
5. Build a small validation set of 10 clips and annotate by hand.

Exit criterion:

Two people can annotate the same 10 clips and understand the ontology without a meeting.

Weeks 2-3: Pilot dataset

1. Select 75 clips from public/research sources and purpose-recorded adult interactions.
2. Annotate cue/access/response/evidence.
3. Run inter-annotator agreement.
4. Revise labels that produce systematic confusion.
5. Create a 25-clip adjudicated dev/test split.

Exit criterion:

At least 200 InteractionEpisodes with acceptable agreement or explicit ambiguity labels.

Weeks 4-5: Baselines

1. Implement flat VLM baseline.
2. Implement schema-prompted JSON baseline.
3. Implement transcript-enhanced baseline.
4. Implement sampled-frame plus transcript baseline.
5. Run at least two strong open VLMs and one frontier model if policy/licensing permits.

Exit criterion:

A table showing where VLMs fail by component, not just a single accuracy number.

Weeks 6-8: CARE-Graph

1. Implement candidate event proposal.
2. Build evidence graph data structures.
3. Train edge scorers on pilot labels.
4. Implement constrained decoding.
5. Implement claim verifier.
6. Run ablations.

Exit criterion:

Full CARE-Graph beats schema-prompted VLM on cue-response pair F1, access macro-F1, and overclaim rate.

Weeks 9-11: Scale dataset

1. Expand to 300-500 clips.
2. Over-sample hard strata: ambiguous recipient, one-modality access, no response, simultaneous cues.
3. Adjudicate test split.
4. Lock the benchmark.
5. Re-run all baselines and ablations.

Exit criterion:

1,000+ InteractionEpisodes, domain-stratified split, stable metric definitions.

Weeks 12-14: Paper

1. Write introduction around the observable-evidence fault line.
2. Put failure examples before method details.
3. Keep related work tight and comparative.
4. Put all ethics and release constraints in the main paper, not only appendix.
5. Create qualitative figures showing:
 - a fluent wrong caption
 - a correct InteractionEpisode graph

- an overclaim rejected by the verifier
- an ambiguous case handled with uncertainty

Exit criterion:

A skeptical reviewer can identify the task, novelty, data, method, metrics, and release story in the first five pages.

13. Paper Outline

1. Introduction

- Captions vs Interaction Episodes
- Evidence-licensed social reasoning
- Contributions

1. Related Work

- social video benchmarks
- grounded reasoning traces
- long-video agentic reasoning
- scene/event graphs
- hallucination and overclaim control

1. Task Definition

- participants
- cues
- access
- responses
- evidence
- claim levels

1. CARE-Bench

- data sources
- purpose-collected clips
- annotation protocol
- agreement
- splits
- ethics and release

1. CARE-Graph

- candidate proposals
- evidence graph
- edge scorer
- constrained decoding

- verifier

1. Experiments

- main results
- ablations
- oracle analysis
- ambiguity and calibration
- overclaim control

1. Discussion

- what VLMs fail at
- what graphs buy
- what remains hard
- why hidden mental states remain out of scope

1. Limitations and Ethics

- not diagnostic
- not attention/comprehension inference
- privacy and surveillance risk
- cultural/contextual bias
- annotation ambiguity

1. Reproducibility

- data card
- annotation guidelines
- evaluation scripts
- prompts
- model outputs
- baseline code
- compute and cost

14. Reviewer-Proof Abstract

Current video-language models can generate fluent descriptions of social scenes, but it remains unclear whether they recover the observable interaction mechanics that license social interpretation. We introduce CARE-Bench, a benchmark for cue-access-response reasoning in naturalistic social videos. CARE formalizes participant-centered InteractionEpisode extraction: identifying who produced a cue, who had observable access to it, whether an observable response followed, when it occurred, which evidence supports the claim, and which mental-state, trait, or clinical claims are unsupported. We annotate naturalistic adult social interactions with cue events, recipient access labels, response events, latency, evidence spans, ambiguity labels, and overclaim categories. We further propose CARE-Graph, a constrained evidence-graph inference system with trainable cue-response and access edge scoring plus claim verification. Across strong VLM, transcript-enhanced, oracle-assisted, and agentic baselines, current models remain weak at access reasoning, cue-response linking, and overclaim control. CARE-

Graph improves structured episode recovery and reduces unsupported claims while preserving useful evidence-grounded explanations. CARE-Bench provides an evidence-first foundation for social video reasoning without requiring hidden mental-state or diagnostic labels.

15. Minimum Submission Checklist

The paper is not ICLR-ready until these are done:

- ☐ frozen schema and metric definitions
- ☐ public annotation guidelines
- ☐ 300-500 clips
- ☐ 1,000+ InteractionEpisodes
- ☐ purpose-collected adult interaction subset
- ☐ adjudicated test split
- ☐ inter-annotator agreement
- ☐ strong open VLM baselines
- ☐ at least one frontier-model baseline if allowed
- ☐ schema-prompted baseline
- ☐ oracle participant and oracle segmentation baselines
- ☐ CARE-Graph full method
- ☐ edge scorer ablation
- ☐ constrained decoder ablation
- ☐ verifier ablation
- ☐ overclaim analysis
- ☐ ambiguity calibration analysis
- ☐ qualitative failure examples
- ☐ ethics and release statement
- ☐ evaluation code
- ☐ dataset card

16. Final Risk Register

Risk: Dataset too small.

Mitigation: Submit only after 1,000+ InteractionEpisodes and stable strata.

Risk: Reviewers say it is "just annotation."

Mitigation: CARE-Graph with trainable edge scoring and constrained decoding must produce real gains.

Risk: Reviewers say social reasoning already exists.

Mitigation: Emphasize observable access/response mechanics and overclaim legality, not generic social QA.

Risk: VLM baselines are weak.

Mitigation: Use strong current open and frontier models, and include oracle-assisted settings.

Risk: Ethical concerns dominate.

Mitigation: Adult-only main dataset, consent or research-permissive sources, no diagnosis, no trait labels, explicit misuse discussion.

Risk: Graph constraints reduce overclaiming by becoming vague.

Mitigation: Report informativeness retention and human preference.

Risk: Annotators disagree.

Mitigation: Treat disagreement as signal; add ambiguity labels and calibration metrics.

Final Standard

The submission should survive this reviewer question:

If I remove every attractive word about agents, social intelligence, and future clinical relevance, what hard technical contribution remains?

The answer must be:

A formal task, benchmark, metric suite, and constrained inference method for evidence-licensed cue-access-response reasoning in social videos, showing that current VLMs fail at observable interaction mechanics and that access-aware graph inference improves the result.

That is the ICLR paper.